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(54) **WATERWAY INSTALLATION STRUCTURE
IN A TOILET**

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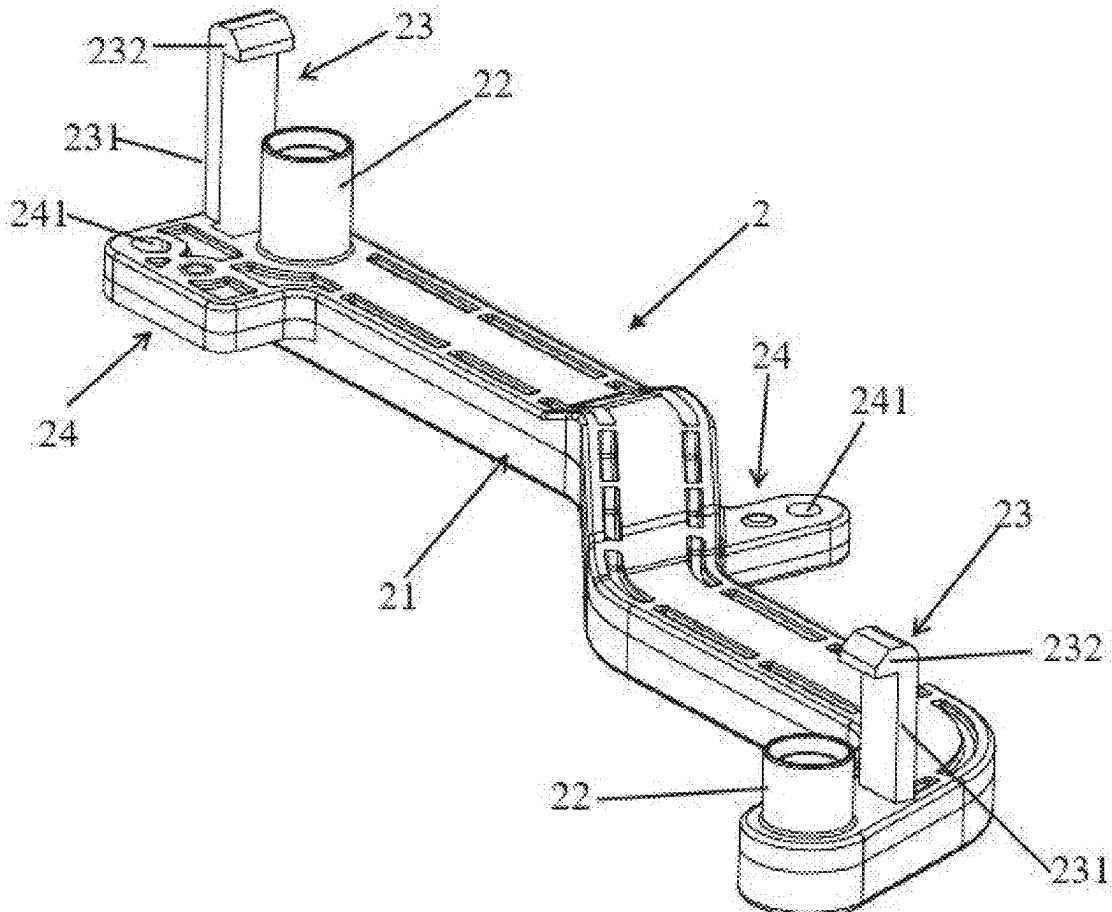
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(57)

ABSTRACT

The application discloses a waterway installation structure in a toilet, comprising at least two waterway modules and a hard connection module for connecting two the waterway modules; each of the waterway modules comprises a waterway joint; the hard connection module comprises a module body having a watercourse and at least two connection pipe joints provided on the module body, each of the connection fittings is in communication with the watercourse; each of the waterway joints on the waterway module is connected and sealed with one connection pipe joint; the waterway joint is plugged into the connection pipe joint; or, the connection pipe joint is plugged into the waterway joint. The waterway installation structure in the toilet disclosed by the application is easy to assemble, and conducive to saving installation space and improving the structural stability.



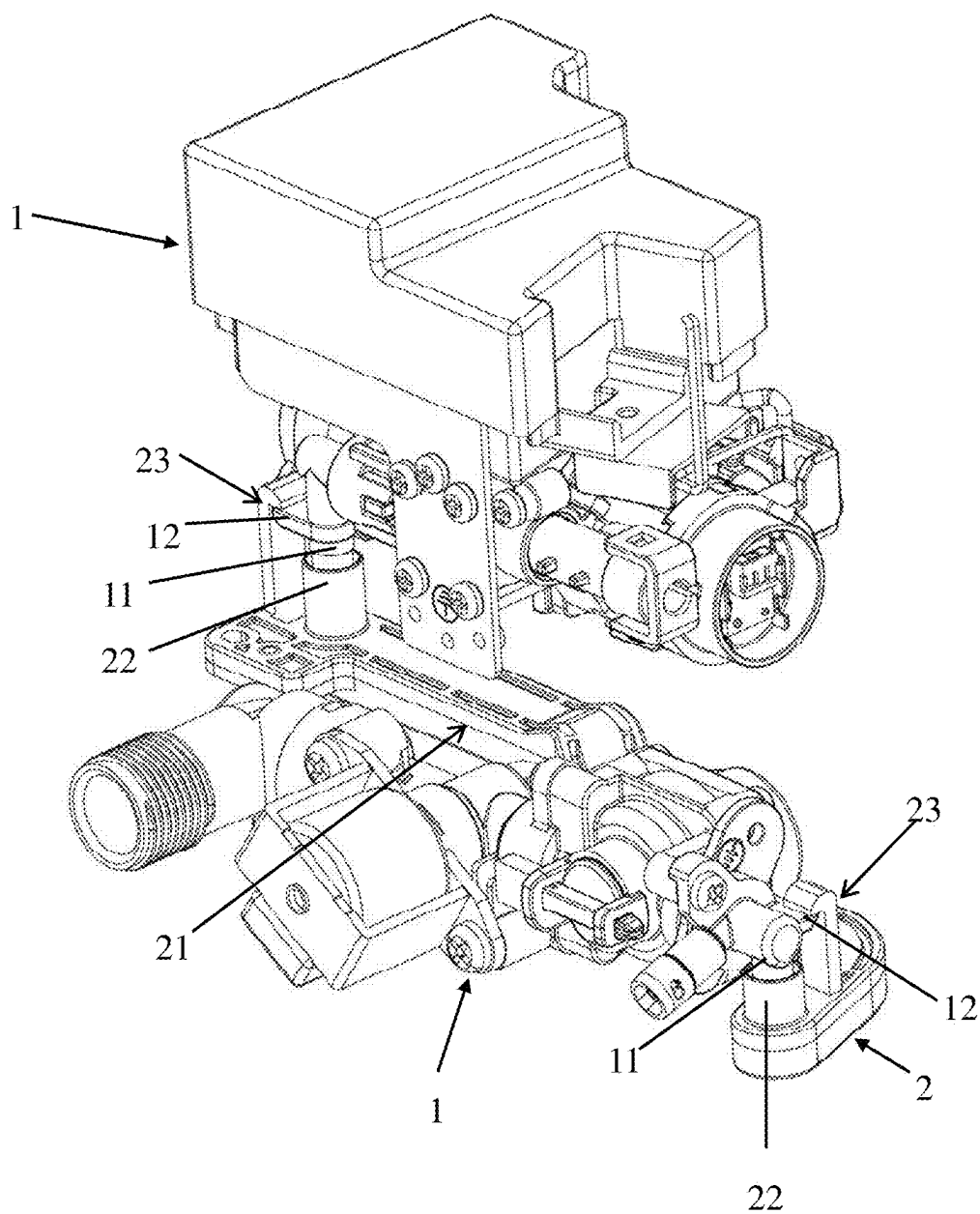


FIG. 1

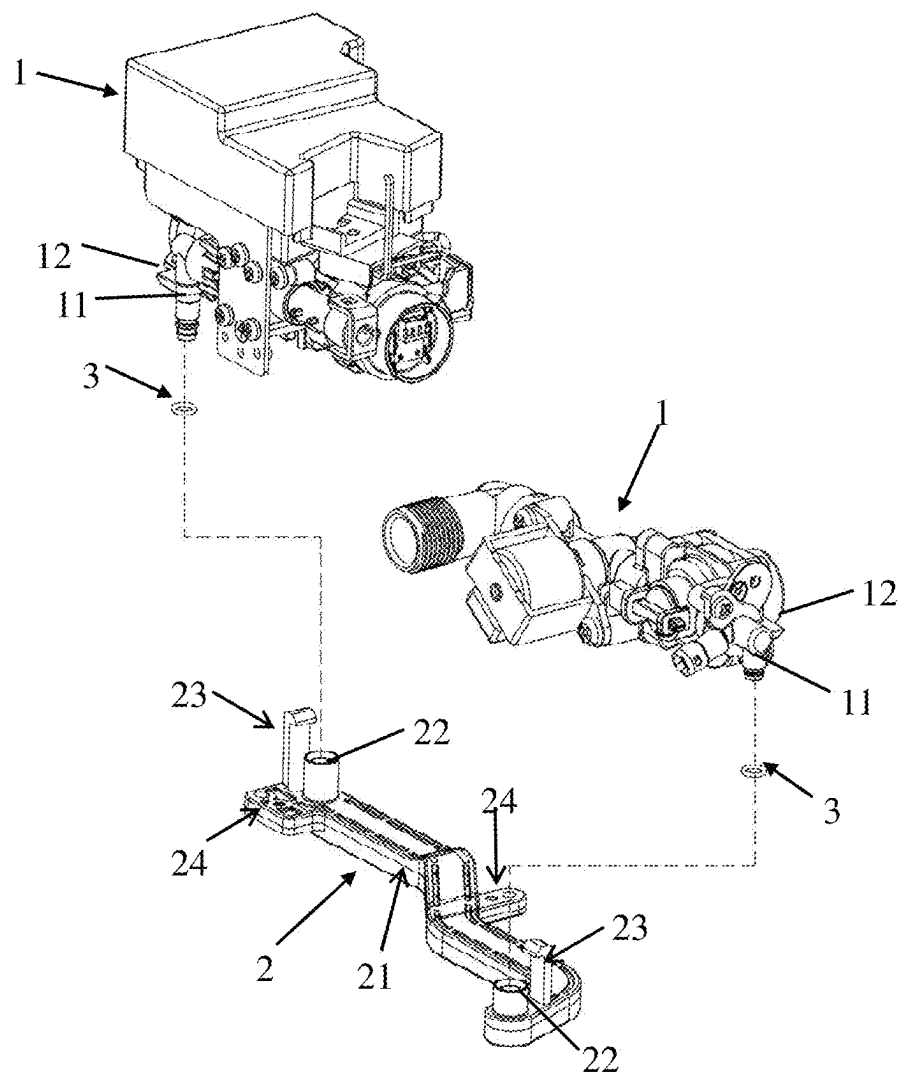


FIG. 2

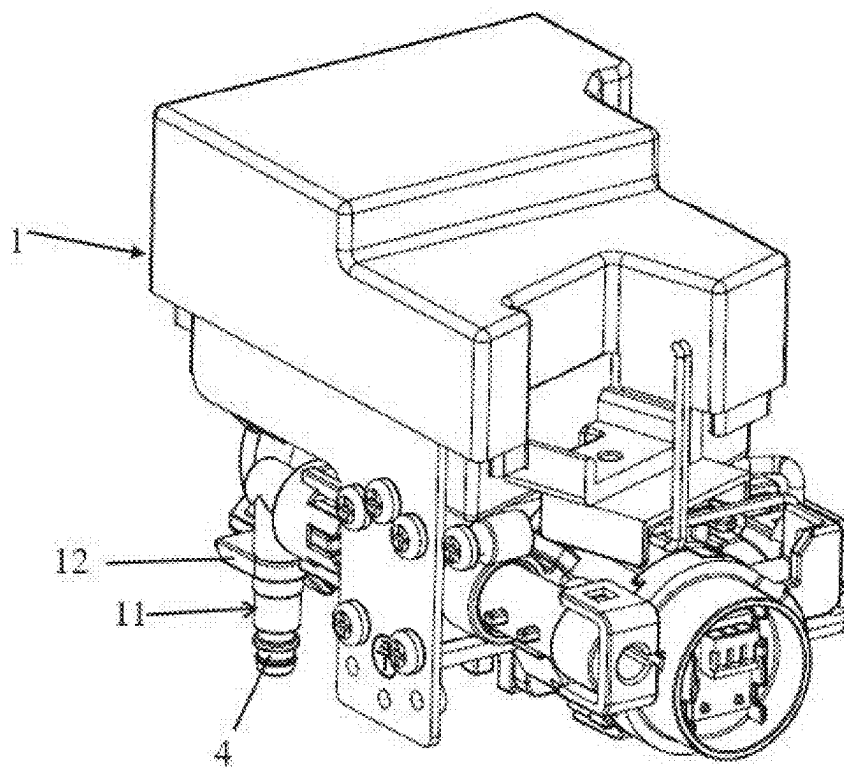


FIG. 3

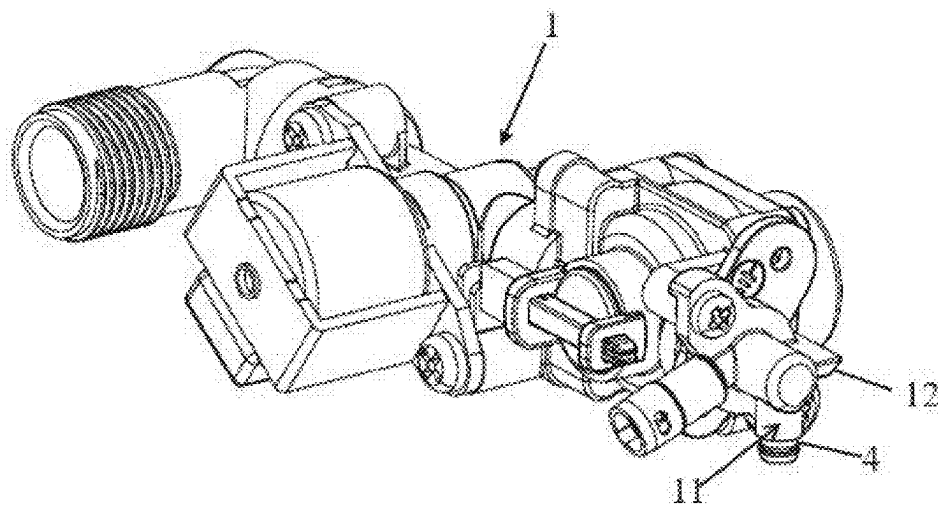


FIG. 4

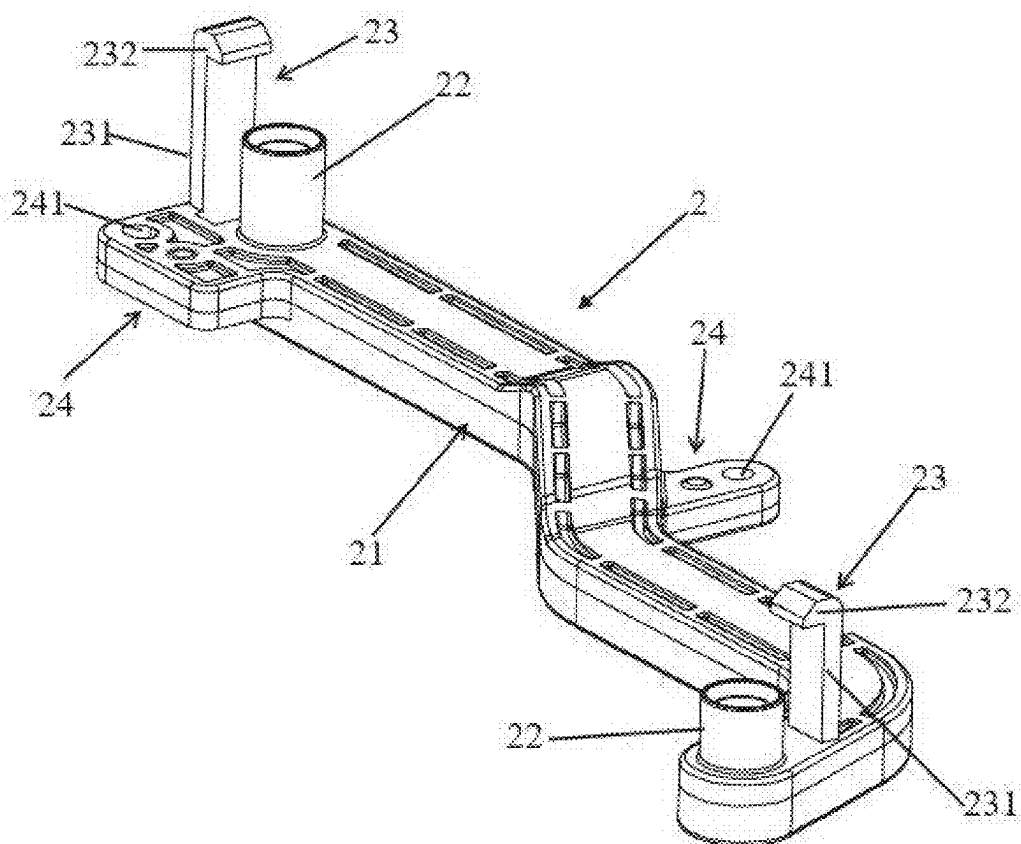


FIG. 5

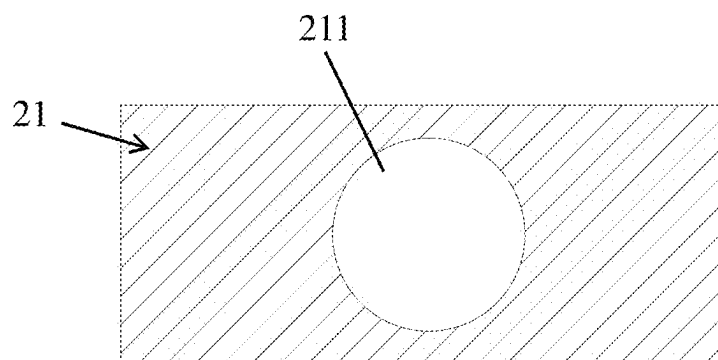


FIG. 6

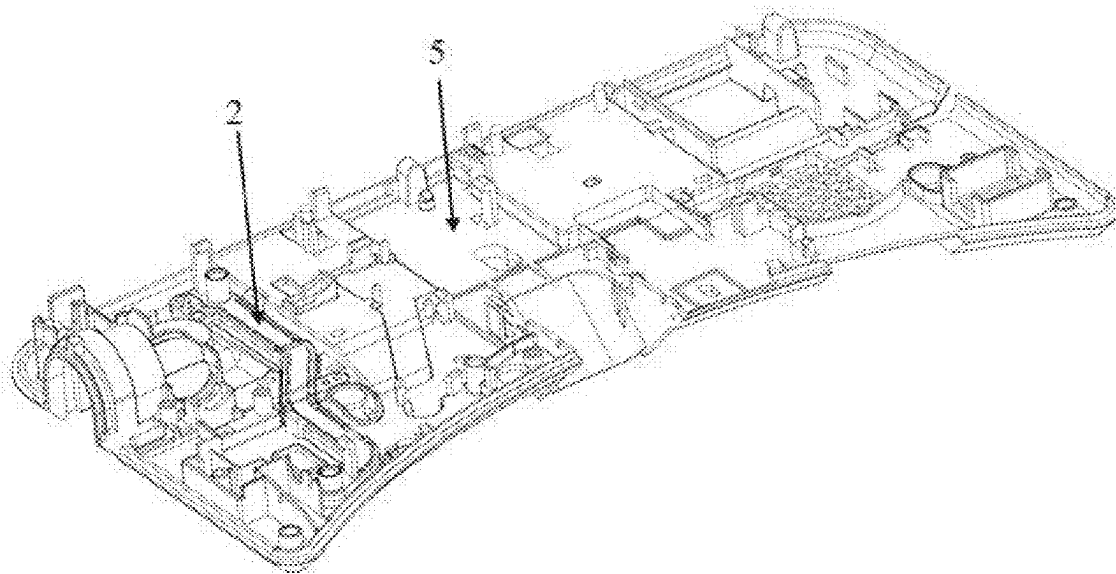


FIG. 7

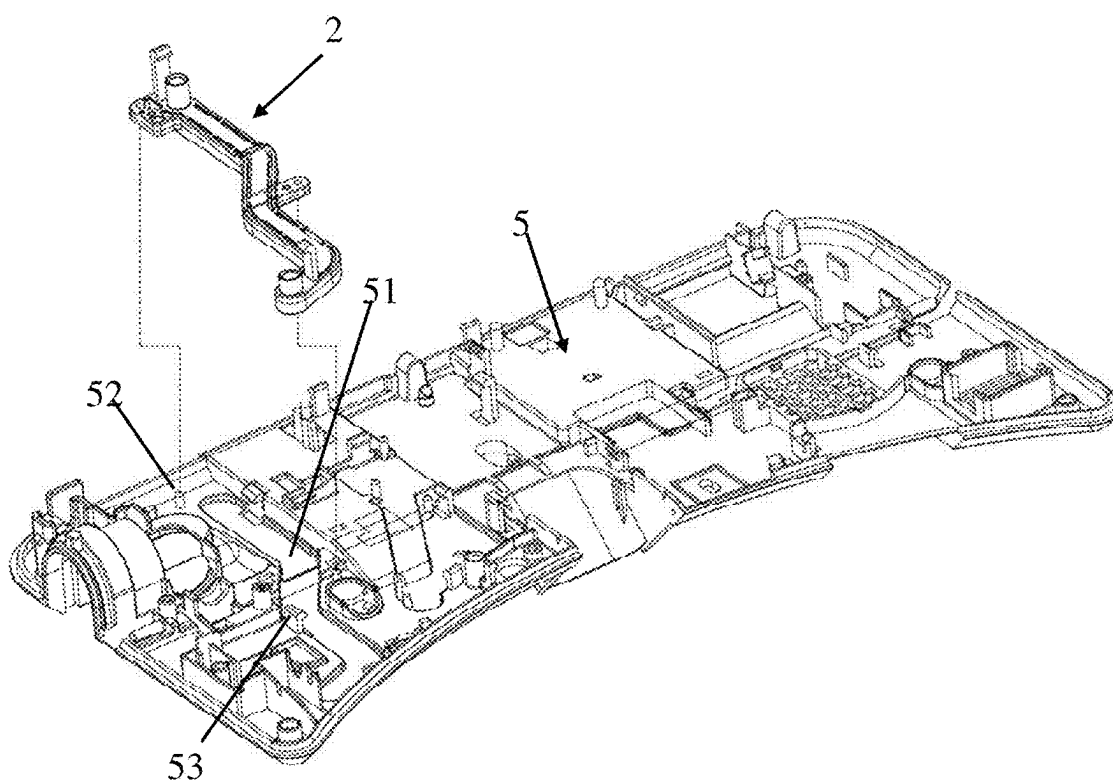


FIG. 8

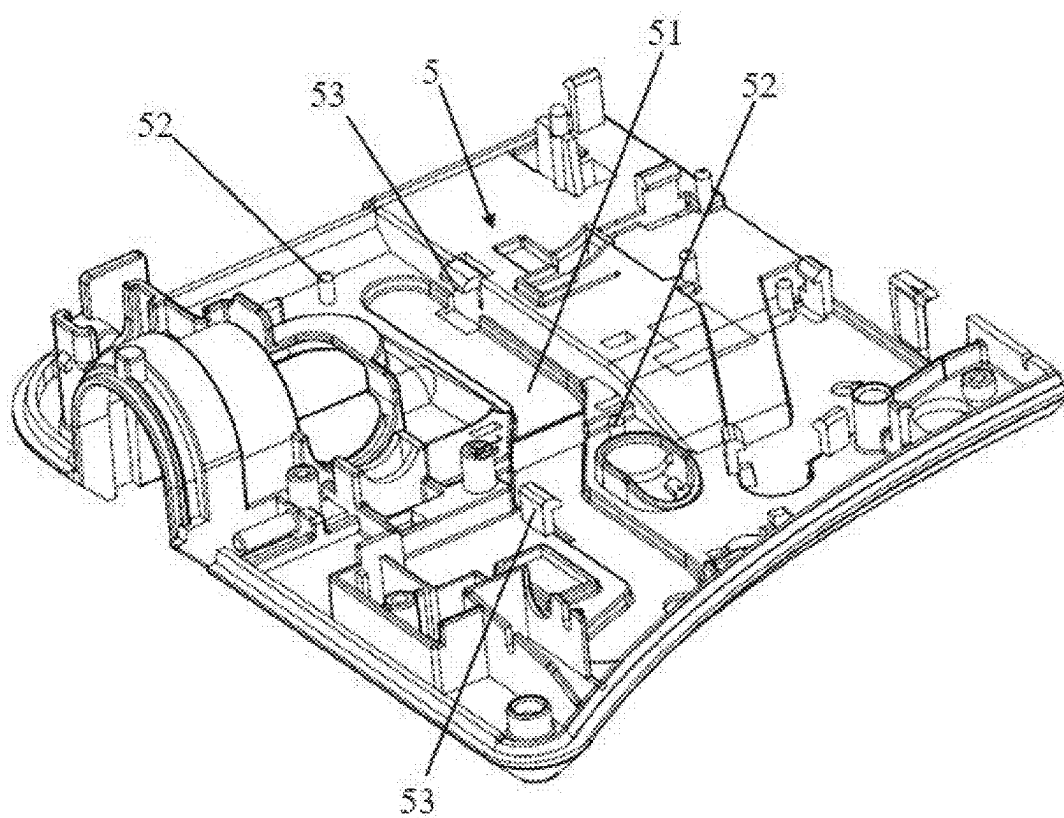


FIG. 9

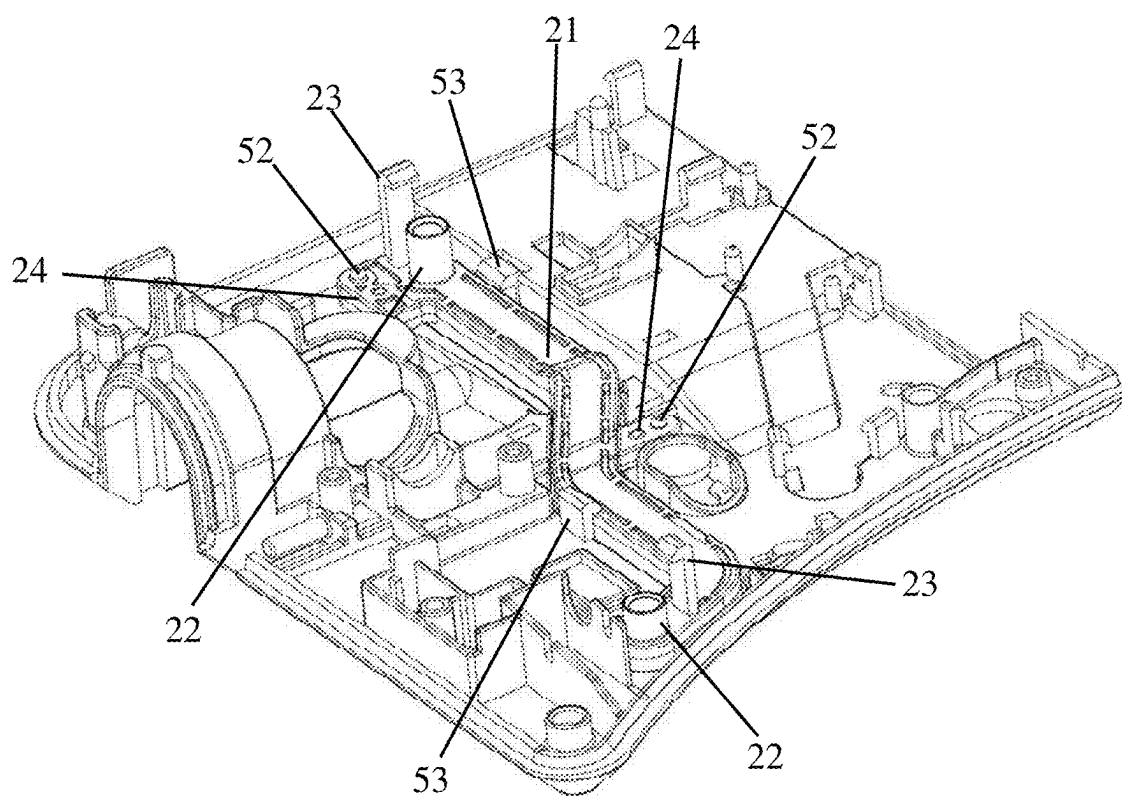


FIG. 10

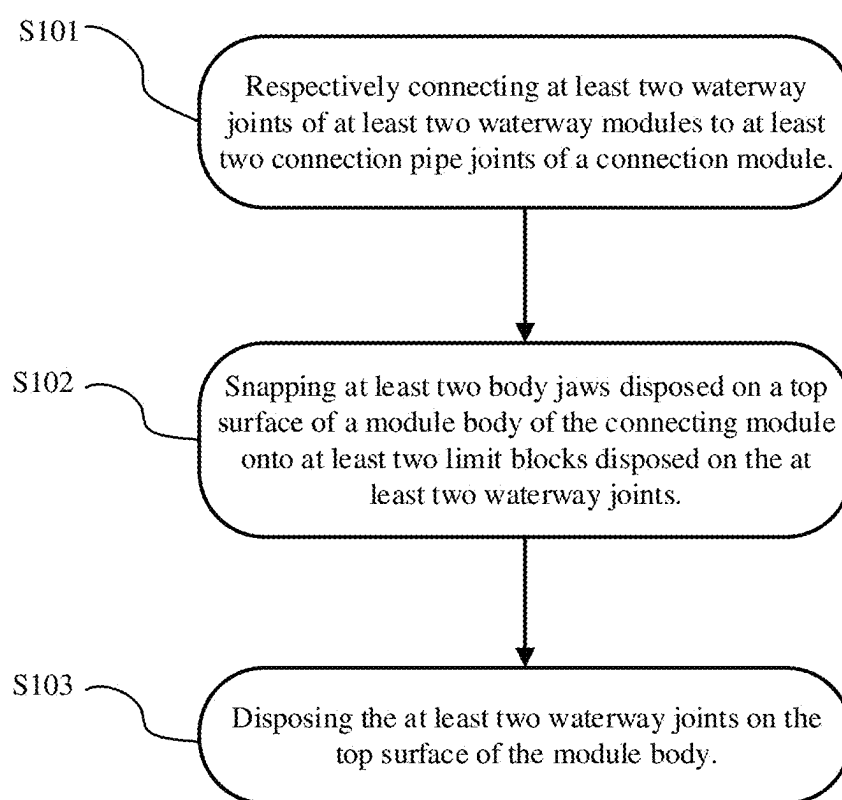


FIG. 11

WATERWAY INSTALLATION STRUCTURE IN A TOILET

CROSS REFERENCE TO RELATED APPLICATION

[0001] This application is a continuation of U.S. patent application Ser. No. 18/244,033 (Docket No. 010222-23043A) filed on Sep. 8, 2023, which is hereby incorporated by reference in its entirety, and which claims the benefit of priority to Chinese Patent Application No. 202222448019.5 filed in the Chinese Intellectual Property Office on Sep. 15, 2022, which is hereby incorporated by reference in its entirety.

FIELD

[0002] The present disclosure relates to the technical field of toilet, and particularly relates to a waterway installation structure in a toilet.

BACKGROUND

[0003] Many waterway modules may be mounted in a back base of a smart toilet, for example, a water purification module, a spray pipe module, a waterway switching module, etc. These modules need to be connected to each other through a water pipe in order to achieve waterway connection.

[0004] The waterway modules that need to be connected to each other are connected through a silicone hose. A certain amount of interference should be left when a silicone hose is cooperated or engaged with a joint, and the hose needs to be used in conjunction with a clamp in the process of assembly to achieve a sealed connection.

[0005] Generally, installation spaces of the waterway modules are small, and the user's hand is difficult to reach into the installation space to conduct the installation. The installation method by using the hose and the clamp is time-consuming and laborious, and the working hour is long. A sufficient bending space needs to be reserved at the connection of the hose to accommodate extra hoses. This takes up a lot of space.

SUMMARY

[0006] The present disclosure aims to overcome the above defects and provides a novelty waterway installation structure in a toilet, which comprises a connection module that is easy to be assembled and takes up small space.

[0007] An embodiment of the present disclosure provides a waterway installation structure in a toilet, comprising at least two waterway modules and a connection module (e.g., a hard connection module or a connection module made from hard material) configured to be connected to the two waterway modules.

[0008] Each of the waterway modules comprises a waterway joint.

[0009] The connection module comprises a module body comprising a watercourse and comprises at least two connection pipe joints disposed on the module body. Each of the connection pipe joints is in communication with (e.g., in fluid communication with) the watercourse.

[0010] The waterway joint of each waterway module is connected to and sealed with one connection pipe joint of the at least two connection pipe joints.

[0011] The waterway joint is plugged (e.g., inserted) into the connection pipe joint, or the connection pipe joint is plugged (e.g., inserted) into the waterway joint.

[0012] In another embodiment of the present disclosure, a seal ring is disposed between the waterway joint and the connection pipe joint.

[0013] In another embodiment of the present disclosure, the waterway joint or the connection pipe joint comprises a limit groove, and the seal ring is assembled or disposed in the limit groove.

[0014] In another embodiment of the present disclosure, the connection pipe joint is formed integrally with the module body.

[0015] In another embodiment of the present disclosure, the module body is provided with a body jaw disposed on a side of each connection pipe joint,

[0016] The waterway module comprises a limit block disposed on a side of each waterway joint.

[0017] The body jaw is in snap-fit connection with the limit block.

[0018] In another embodiment of the present disclosure, the waterway installation structure further comprises a structural base, and the structural base comprises a mounting groove configured to mount the module body.

[0019] The module body is removably mounted in the mounting groove.

[0020] In another embodiment of the present disclosure, the structural base is provided with at least one positioning pin around the mounting groove. Correspondingly, the module body is provided with at least one positioning lug plate.

[0021] Each of the at least one positioning lug plate comprises a positioning hole configured to cooperate with the at least one positioning pin.

[0022] The positioning lug plate is disposed on an outside of the mounting groove, and each of the at least one positioning pin passes through one corresponding positioning hole with a clearance therebetween.

[0023] In another embodiment of the present disclosure, the mounting groove is provided with the at least one positioning pin disposed on each of opposite sides of the mounting groove. Correspondingly, the module body is provided with the at least one positioning lug plate disposed on each of opposite sides of the module body.

[0024] In another embodiment of the present disclosure, the structural base is provided with at least one base jaw around the mounting groove, and the base jaw is in snap-fit connection with the module body.

[0025] In another embodiment of the present disclosure, the mounting groove is provided with the at least one base jaw disposed on each of opposite sides of the mounting groove. Each of the base jaws is in snap-fit connection with the module body.

[0026] By adopting the technical solutions above, the present disclosure has the following beneficial effects:

[0027] According to the waterway installation structure in a toilet provided by the present disclosure, the connection module is adopted, the waterway module comprises the waterway joint, and the connection module comprises the connection pipe joint. When the waterway installation structure is connected or assembled, the waterway joint may be directly inserted into the connection pipe joint to achieve a sealed connection, or the connection pipe joint may be directly inserted into the waterway joint to achieve a sealed connection. It is easy to assemble the waterway installation

structure, and thus time and effort may be saved. The connection module does not take up extra installation space but takes up small space. This is convenient for structural arrangement. The connection module plays a structural connection role between the two waterway modules and helps to improve the stability of the connection between the two waterway modules.

[0028] In summary, the waterway installation structure in the toilet provided by the present disclosure is easy to be assembled, saves the installation space, and also improves the structural stability.

BRIEF DESCRIPTION OF THE FIGURES

[0029] With reference to the drawings, the contents disclosed by the present disclosure should be more easily understood. It should be understood that these drawings are merely used for illustration and are not intended to limit the protection scope of the present disclosure. In the drawings:

[0030] FIG. 1 is a perspective view of a waterway installation structure in the toilet according to an example of the present disclosure;

[0031] FIG. 2 is an exploded view of the waterway installation structure in the toilet shown in FIG. 1;

[0032] FIG. 3 is a perspective view of a waterway module of a structure;

[0033] FIG. 4 is a perspective view of a waterway module of another structure;

[0034] FIG. 5 is a perspective view of a connection module;

[0035] FIG. 6 is a sectional view of a module body with a watercourse;

[0036] FIG. 7 is a perspective view of the connection module connected to a structural base;

[0037] FIG. 8 is an exploded view of the connection module connected to the structural base;

[0038] FIG. 9 is a partially enlarged view of the structural base;

[0039] FIG. 10 is an enlarged view of the connection module connected to the structural base; and

[0040] FIG. 11 is a flow chart of a method for assembling the waterway installation structure in the toilet according to an example of the present disclosure.

DETAILED DESCRIPTION

[0041] The specific embodiments of the present disclosure are further described with reference to the drawings hereinafter. Same or equivalent parts are denoted by same reference numerals. It should be noted that the terms “front”, “back”, “left”, “right”, “up” and “down” used in the following description refer to the directions in the drawings, and the terms “inner” and “outer” refer to the directions towards or far away from geometric centers of specific parts respectively.

[0042] FIG. 1 is a perspective view of a waterway installation structure in the toilet according to an example of the present disclosure. FIG. 2 is an exploded view of the waterway installation structure in the toilet shown in FIG. 1. FIG. 3 is a perspective view of a waterway module of a structure. FIG. 4 is a perspective view of a waterway module of another structure. FIG. 5 is a perspective view of a connection module. FIG. 6 is a sectional view of a module body with a watercourse. As shown in FIGS. 1-6, a waterway installation structure in a toilet according to an embodi-

ment of the present disclosure comprises at least two waterway modules 1 and a hard connection module 2 configured to be connected to the at least two waterway modules 1.

[0043] Each waterway module 1 comprises a waterway joint 11.

[0044] The hard connection module 2 comprises a module body 21 comprising a watercourse 211 and comprises at least two connection pipe joints 22 disposed on the module body 21. Each of the connection pipe joints 22 is in communication with (e.g., in fluid communication with) the watercourse 211.

[0045] The waterway joint 11 on each waterway module 1 is connected to and sealed with one connection pipe joint 22.

[0046] The waterway joint 11 is plugged (e.g., inserted) into the connection pipe joint 22, or the connection pipe joint 22 is plugged (e.g., inserted) into the waterway joint 11.

[0047] The present disclosure provides a waterway installation structure mainly used in the toilet. Specifically, the waterway installation structure may be mounted in a back base of the toilet or a water tank. The back base used in this present disclosure may be a mounting base of a toilet cover.

[0048] The waterway installation structure comprises the waterway module 1 and the hard connection module 2. The waterway module 1 may be a water-using device, such as a motor, a water purifier, a spray pipe, etc. The waterway module 1 may also be a water-passing device, such as a water valve, pipe fitting, etc. The waterway module 1 may also be a combination of the water-using devices and the water-passing devices.

[0049] The waterway installation structure comprises at least two waterway modules 1. Each waterway module 1 comprises a waterway joint 11. The waterway joint 11 may be an inlet or outlet of the waterway module 1, and specific settings may be made according to actual needs. When the at least two waterway modules 1 are connected to a connection module 2 (e.g., a hard connection module or a connection module made from hard material). One waterway joint 11 of one of the at least two waterway modules 1 is an outlet of the corresponding waterway module 1, a waterway joint 11 of the other waterway module 1 of the at least two waterway modules 1 is an inlet of the corresponding waterway module 1.

[0050] When two waterway modules 1 need to achieve waterway connection, the hard connection module 2 may be used for the connection.

[0051] The hard connection module 2 may be a plastic connection module or a metal connection module. Compared to the rubber hose, the plastic connection module or the metal connection module is harder in texture and will not bend. Thus, the hard connection module 2 may be called “hard connection module.”

[0052] The hard connection module 2 comprises a module body 21 and at least two connection pipe joints 22. As needed, the module body 21 may be configured in an elongated shape with a watercourse 211 inside. The number of the connection pipe joints 22 is configured according to the number of waterway joints 11 to be connected. The at least two connection pipe joints 22 are connected to the module body 21, and each connection pipe joint 22 is in communication with (e.g., in fluid communication with) the watercourse 211.

[0053] In this present disclosure, both the waterway joint 11 and the connection pipe joint 22 are cylindrical.

[0054] According to actual needs, an outer diameter of the waterway joint 11 may be set to be slightly smaller than an inner diameter of the connection pipe joint 22, so that the waterway joint 11 can be directly plugged into the connection pipe joint 22 to achieve the connection.

[0055] According to actual needs, alternatively the outer diameter of the connection pipe joint 22 may be set to be slightly smaller than the inner diameter of the waterway joint 11, so that the connection pipe joint 22 can be directly plugged into the waterway joint 11 to achieve the connection.

[0056] When the waterway installation structure is assembled, the waterway joint 11 on each waterway module 1 may be directly plugged into the corresponding connection pipe joint 22 to achieve a sealed connection, or each connection pipe joint 22 of the hard connection module 2 may be plugged into the corresponding waterway joint 11 to achieve a sealed connection.

[0057] As needed, a tapered sealing surface is provided on surfaces (e.g., mating surfaces) of the at least two waterway joints 11 and the at least two connection pipe joints 22 to achieve automatic sealing after the waterway joint 11 is plugged into the connection pipe joint 22 or the connection pipe joint 22 is plugged into the waterway joint 11.

[0058] It is also possible to coat sealant onto the mating surfaces of the waterway joint 11 and the connection pipe joint 22, as needed, to achieve automatic sealing after the waterway joint 11 is plugged into the connection pipe joint 22 or the connection pipe joint 22 is plugged into the waterway joint 11.

[0059] The waterway installation structure in the toilet provided in the present disclosure adopts the hard connection module 2. When the waterway installation structure is connected or assembled, the waterway joint 11 may be directly inserted into the connection pipe joint 22 to achieve a sealed connection, or the connection pipe joint 22 may be directly inserted into the waterway joint 11 to achieve a sealed connection. It is easy to assemble the waterway installation structure, and thus time and effort may be saved. The hard connection module 2 does not take up extra installation space but takes up small space. This is convenient for structural arrangement. The hard connection module 2 plays a structural connection role between the two waterway modules 1 and helps to improve the stability of the connection between the two waterway modules 1.

[0060] In summary, the waterway installation structure in the toilet provided by the present disclosure is easy to be assembled, saves the installation space, and also improves the structural stability.

[0061] In an embodiment, as shown in FIG. 2, a seal ring 3 is disposed between the waterway joint 11 and the connection pipe joint 22 to further improve sealing performance. The seal ring 3 is a rubber seal ring.

[0062] In an embodiment, as shown in FIGS. 3 and 4, the waterway joint 11 or the connection pipe joint 22 comprises a limit groove 4, and the seal ring 3 is assembled or disposed in the limit groove 4 to improve the assembly stability of the seal ring 3.

[0063] If the waterway joint 11 is plugged into the connection pipe joint 22, the limit groove 4 is configured or disposed on an outer periphery surface of the waterway joint 11.

[0064] If the connection pipe joint 22 is plugged into the waterway joint 11, the restriction groove 4 is configured or disposed on an outer peripheral surface of the connection pipe joint 22.

[0065] In an embodiment, the connection pipe joint 22 is formed integrally with the module body 21, resulting in high structural strength. If the hard connection module 2 is a hard plastic component, the connection pipe joint 22 is formed integrally with the module body 21 by injection molding. If the hard connection module 2 is a metal component, the connection pipe joint 22 is formed integrally with the module body 21 by casting.

[0066] In an embodiment, as shown in FIGS. 1-5, the module body 21 is provided with a body jaw 23 disposed on a side of each connection pipe joint 22.

[0067] The waterway module 1 comprises a limit block 12 disposed on a side of the waterway joint 11.

[0068] The body jaw 23 is in snap-fit connection with the limit block 12.

[0069] In this embodiment, the hard connection module 2 comprises the body jaw 23 to provide a limit to the module body 21. The body jaw 23 is connected to the module body 21, and the body jaw 23 and the module body 21 may be formed integrally. The body jaw 23 extends from the module body 21 toward the waterway module 1.

[0070] Specifically, one body jaw 23 is disposed on the side of each connection pipe joint 22.

[0071] Each waterway module 1 comprises one limit block 12, and the limit block 12 extends horizontally or laterally. The limit block 12 may be formed integrally with the waterway module 1. The limit block 12 is disposed on the side of the waterway joint 11.

[0072] When the waterway installation structure is assembled, each body jaw 23 on the hard connection module 2 is snapped onto limit block 12 of one waterway module 1 to improve the assembly stability of the hard connection module 2 and the waterway module 1.

[0073] In an embodiment, the hard connection module 2 is disposed under or below the waterway module 1, and the body jaw 23 is disposed on a top surface of the module body 21.

[0074] Specifically, the body jaw 23 comprises a standing plate 231 connected to the module body 21 and a hanging plate 232 connected to a top of the standing plate 231 and extending toward a side of the standing plate 231. The hanging plate 232 is hung on the limit block 12 when the hard connection module 2 is assembled to the waterway module 1.

[0075] As needed, a top surface of the hanging plate 232 and a bottom surface of the limit block 12 each may be configured with or comprise a corresponding bevel. When the hard connection module 2 is assembled with the waterway module 1, the standing plate 231 is deformed under the action of the bevel, so that the hanging plate 232 may move onto the limit block 12, and thus the hanging plate 232 may automatically hang on the limit block 12 after the reset of the standing plate 231.

[0076] FIG. 7 is a perspective view of the connection module connected to a structural base. FIG. 8 is an exploded view of the connection module connected to the structural base. FIG. 9 is a partially enlarged view of the structural base. FIG. 10 is an enlarged view of the connection module connected to the structural base. In an embodiment, as shown in FIGS. 7-10, the waterway installation structure

also comprises a structural base **5** comprising a mounting groove **51** configured to mount the module body **21**. The module body **21** is removably mounted in the mounting groove **51**.

[0077] In this embodiment, the waterway installation structure comprises the structural base **5**, and the structural base **5** may be a pedestal of a back base of a toilet or toilet bidet.

[0078] The structural base **5** has the mounting groove **51**, which matches the shape contour of the module body **21**, and the module body **21** is removably mounted in the mounting groove **51**, enhancing the installation stability of the hard connection module **2**.

[0079] In this present disclosure, the at least two hard connection modules **2** have installation positions at a top surface of the structural base **5**, and the module body **21** is installed in the mounting groove **51** of the structural base **5** to further improve the stability of the waterway installation structure.

[0080] In an embodiment, as shown in FIGS. 7-10, at least one positioning pin **52** is disposed configured on the structural base **5** around the mounting groove **51**. Correspondingly, at least one positioning lug plate **24** is disposed on the module body **21**.

[0081] Each of the positioning lug plates **24** comprises a positioning hole **241** configured to cooperate with (e.g., receive) the positioning pin **52**.

[0082] The positioning lug plate **24** is disposed on an outside of the mounting groove **51**, and each of the positioning pins **52** passes through the one positioning hole **241** with a clearance therebetween.

[0083] In this embodiment, at least one positioning pin **52** is disposed on the structural base **5**, and the positioning pin **52** is disposed on the outside of the mounting groove **51**. The module body **21** is correspondingly provided with at least one positioning lug plate **24**. The positioning lug plate **24** extends laterally from the module body **21**, which may be formed integrally with the module body **21**. Each of the positioning lugs **24** has at least one positioning hole **241**, which is used to cooperate with the positioning pin **52** to achieve the positioning.

[0084] When the waterway installation structure is assembled, the positioning lug plate **24** is disposed on the outside of the mounting groove **51**, and the positioning lug plate **24** covers the positioning pin **52**. Each positioning pin **52** passes through and is in clearance fit with one positioning hole **241**.

[0085] In an embodiment, as shown in FIGS. 7-10, at least one positioning pin **52** is disposed on each of the opposite sides of the mounting groove **51** respectively. Correspondingly, at least one positioning lug plate **24** is disposed on each of the opposite sides of the module body **21** respectively.

[0086] In this embodiment, the positioning pin **52** is disposed on each side of the long side of the mounting groove **51** respectively. Correspondingly, the positioning lug plate **24** is disposed on each of the two long sides of the module body **21** respectively. When the waterway installation structure is assembled, the positioning lug plate **24** is disposed on the outside of the mounting groove **51**, and the positioning lug plate **24** covers the positioning pin **52**. Each positioning pin **52** cooperates with the positioning hole **241** on the corresponding positioning lug plate **24** to achieve the positioning and improve the structural stability.

[0087] In an embodiment, as shown in FIGS. 7-10, the structural base **5** is provided with at least one base jaw **53** around the mounting groove **51**, and the base jaw **53** is in snap-fit connection with the module body **21**.

[0088] In this embodiment, the base jaw **53** is provided on the structural base **5**, and the base jaw **53** is on the outside of the mounting groove **51**, and the mounting groove **51** may also be provided directly on a groove wall of the mounting groove **51**. The base jaw **53** may have a similar structural design to the body jaw **23**.

[0089] After the module body **21** is assembled into the mounting groove **51**, the snap plate or hanging plate of the base jaw **53** is snapped onto the top surface of the module body **21**. Thus, the module body **21** is limited in the mounting groove **51**, and the assembly stability of the module body **21** to the mounting groove **51** may be improved.

[0090] In an embodiment, as shown in FIGS. 7-10, the mounting groove **51** comprises at least one base jaw **53** disposed on each opposite sides respectively, and each of the base jaws **53** may be snapped onto the module body **21**.

[0091] In this embodiment, the base jaw **53** is provided on each side of the long side of the mounting groove **51**. During the assembly, the base jaws **53** on both sides are snapped onto the module body **21**, and thus the assembly stability of the module body **21** may be further improved.

[0092] In an embodiment, the toilet includes a base (e.g., a pedestal, bowl, etc.) and a tank. The base is configured to be attached to another object such as a drainpipe, floor, or another suitable object. The base includes a bowl, a sump (e.g., a receptacle) disposed below the bowl, and a trapway fluidly connecting the bowl to a drainpipe or sewage line. The tank may be supported by the base, such as an upper surface of a rim. The tank may be integrally formed with the base as a single unitary body. In other embodiments, the tank may be formed separately from the base and coupled (e.g., attached, secured, fastened, connected, etc.) to the base. The toilet may further include a tank lid covering an opening and inner cavity in the tank. The toilet may include a seat assembly including a seat and a seat cover rotatably coupled to the base. The toilet may further include a hinge assembly.

[0093] In another embodiment, the toilet may be a tankless toilet. The toilet includes a base and a seat assembly coupled to the base. The base includes a bowl, a sump disposed below the bowl, and a trapway fluidly connecting the bowl to a drainpipe or sewage line. The toilet includes a waterline that supplies the toilet with water. The toilet may further include a seat assembly including a seat and a seat cover rotatably coupled to the base. The toilets described above are provided herein as non-limiting examples of toilets that may be configured to utilize aspects of the present disclosure.

[0094] In some examples, a bidet may be included in a seat or pedestal of a toilet. In other examples, the bidet may be manufactured separately from and attached or coupled to a seat or pedestal of a toilet. The bidet includes a housing. The housing is configured to receive a flow of water through a housing inlet and dispense the flow of water from a housing outlet. The housing inlet and housing outlet may be located on opposite ends of the housing from one another, such that water may flow through the housing from the housing inlet to the housing outlet. In some examples, the housing further includes a chamber. As the housing receives the flow of water, the chamber may fill with water and provide a flow of water between the housing inlet and the housing outlet. The

chamber may be configured to contain the flow of water and direct the flow of water from the housing inlet to the housing outlet. After the chamber has filled with water, the flow of water may travel along a substantially linear path between the housing inlet and the housing outlet. In some examples, one or more walls within the housing may be included to help direct a flow of water between the housing inlet and the housing outlet. The bidet may further include a housing inlet conduit configured to direct a flow of water to the housing inlet. The housing inlet conduit may be coupled to a water supply such as tank or waterline. The housing may further include a gear assembly or a portion of the gear assembly. **[0095]** FIG. 11 is a flow chart of a method for assembling the waterway installation structure in the toilet according to an example of the present disclosure. The the waterway installation structure in the toilet used in the method may be the waterway installation structure in the toilet according to any of the foregoing examples and may be configured to perform an operation, function, or the like as described in the present disclosure.

[0096] At act S101, the at least two waterway joints 11 of the at least two waterway modules 1 are respectively connected to the at least two connection pipe joints 22 of the connection module 2.

[0097] As noted above, the waterway joint 11 is inserted into the corresponding connection pipe joint 22, or the connection pipe joint 22 is inserted into the corresponding waterway joint 11. The seal ring 3 is disposed between the waterway joint 11 and the connection pipe joint 22 to further improve sealing performance. The seal ring 3 is a rubber seal ring. The waterway joint 11 or the connection pipe joint 22 comprises a limit groove 4, and the seal ring 3 is assembled or disposed in the limit groove 4 to improve the assembly stability of the seal ring 3. If the waterway joint 11 is inserted into the connection pipe joint 22, the limit groove 4 is configured or disposed on an outer periphery surface of the waterway joint 11. If the connection pipe joint 22 is inserted into the waterway joint 11, the restriction groove 4 is configured or disposed on an outer peripheral surface of the connection pipe joint 22.

[0098] At act S102, the at least two body jaws 23 disposed on a top surface of the module body 21 of the connecting module 2 are snapped onto the at least two limit blocks 12 disposed on the at least two waterway joints 11.

[0099] As noted above, when the waterway installation structure is assembled, each body jaw 23 on the hard connection module 2 is snapped onto limit block 12 of one waterway module 1 to improve the assembly stability of the hard connection module 2 and the waterway module 1. The body jaw 23 comprises a standing plate 231 connected to the module body 21 and a hanging plate 232 connected to a top of the standing plate 231 and extending toward a side of the standing plate 231. The hanging plate 232 is hung on the limit block 12 when the hard connection module 2 is assembled to the waterway module 1.

[0100] At act S103, the at least two waterway joints 11 are disposed on the top surface of the module body 21.

[0101] As noted above, the hard connection module 2 is disposed under or below the waterway module 1, and the body jaw 23 is disposed on a top surface of the module body 21. As needed, a top surface of the hanging plate 232 and a bottom surface of the limit block 12 each may be configured with or comprise a corresponding bevel. When the hard connection module 2 is assembled with the waterway mod-

ule 1, the standing plate 231 is deformed under the action of the bevel, so that the hanging plate 232 may move onto the limit block 12, and thus the hanging plate 232 may automatically hang on the limit block 12 after the reset of the standing plate 231.

[0102] The above technical solutions may be combined as required to achieve the best technical effect.

[0103] The above are merely the principle and the preferred embodiments of the present disclosure. It should be pointed out that, for those of ordinary skills in the art, several other modifications may be made on the basis of the principle of the present disclosure, which should also be regarded as falling in the protection scope of the present disclosure.

We claim:

1. A waterway installation structure in a toilet, the waterway installation structure comprising:

a connection module mounted to the toilet, the connection module comprising:

a module body including a watercourse and a connection pipe joint in fluid communication with the watercourse; and

at least one waterway module including a waterway joint configured to be connected to the connection pipe joint,

wherein the waterway joint is configured to be inserted into the connection pipe joint or the connection pipe joint is configured to be inserted into the waterway joint.

2. The waterway installation structure of claim 1, further comprising:

a seal ring disposed between the waterway joint and the connection pipe joint.

3. The waterway installation structure of claim 2, further comprising:

a limit groove disposed on the waterway joint or the connection pipe joint, the seal ring disposed in the limit groove.

4. The waterway installation structure of claim 1, wherein the connection pipe joint is formed integrally with the module body.

5. The waterway installation structure of claim 1, further comprising:

a body jaw disposed on the module body on a side of the connection pipe joint; and

a limit block disposed on the waterway joint, wherein the body jaw is configured to be connected to the limit block.

6. The waterway installation structure of claim 5, wherein:

the body jaw includes a standing plate extending from the module body, and a hanging plate connected to the standing plate;

the body jaw is disposed on a top surface of the module body; and

the water module is disposed on the top surface of the module body.

7. The waterway installation structure in the toilet according to claim 1, further comprising:

a structural base comprising a mounting groove,

wherein the module body is removably mounted in the mounting groove.

8. The waterway installation structure in the toilet according to claim **7**, the waterway installation structure further comprising:

at least one positioning pin disposed on the structural base around the mounting groove,
wherein the module body includes at least one positioning lug plate,
wherein the at least one positioning lug plate includes at least one positioning hole configured to receive the at least one positioning pin, and
wherein the positioning lug plate is disposed on an outside of the mounting groove when the at least one positioning pin passes through the at least one positioning hole.

9. The waterway installation structure in the toilet according to claim **8**, wherein:

the at least one positioning pin comprises two positioning pins respectively disposed on opposite sides of the mounting groove, and

the at least one positioning lug plate comprises two positioning lug plates respectively disposed on opposite sides of the module body.

10. The waterway installation structure in the toilet according to claim **7**, further comprising:

at least one base jaw disposed on the structural base and around the mounting groove, the at least one base jaw configured to be connected to the module body.

11. The waterway installation structure in the toilet according to claim **10**, wherein the at least one base jaw comprises two base jaws respectively on opposite sides, the two base jaws configured to be connected to the module body.

12. A method for assembling a waterway installation structure, the method comprising:

connecting at least two waterway joints of at least two waterway modules to respective ones of at least two connection pipe joints of a connection module;
disposing the at least two waterway joints on a top surface of the connection module; and
mounting the waterway installation structure to a toilet.

13. The method of claim **12**, further comprising:

connecting a seal ring between a waterway joint of the at least two waterway joints and a corresponding connection pipe joint of the at least two connection pipe joints.

14. The method of claim **12**, wherein connecting the at least two waterway joints of the at least two waterway modules to respective ones of the at least two connection pipe joints of the connection module includes:

inserting the at least two waterway joints into respective ones of the at least two connection pipe joints.

15. The method of claim **12**, wherein connecting the at least two waterway joints of the at least two waterway modules to respective ones of the at least two connection pipe joints of the connection module includes:

inserting the at least two connection pipe joints into respective ones of the at least two waterway joints.

16. The method of claim **12**, further comprising:

disposing a module body including the at least two connection pipe joints in a mounting groove of a structural base.

17. The method of claim **16**, further comprising:

inserting at least one positioning pin disposed on the structural base into at least one positioning hole of at least one positioning lug plate of the module body.

18. A waterway installation structure in a toilet, the waterway installation structure comprising:

a connection module mounted to the toilet, the connection module comprising:

a module body including a watercourse and a connection pipe joint in fluid communication with the watercourse; and

at least one waterway module including a waterway joint configured to be connected to the connection pipe joint; and

a structural base including a mounting groove, the mounting groove configured to receive the module body, wherein the waterway joint is configured to be inserted into the connection pipe joint or the connection pipe joint is configured to be inserted into the waterway joint.

19. The waterway installation structure of claim **18**, further comprising:

a seal ring disposed between the waterway joint and the connection pipe joint; and

a limit groove disposed on the waterway joint or the connection pipe joint,

wherein the seal ring disposed in the limit groove.

20. The waterway installation structure of claim **18**, further comprising:

a body jaw disposed on the module body on a side of the connection pipe joint; and

a limit block disposed on the waterway joint, wherein the body jaw is configured to be connected to the limit block.

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